LO1: Requirements

LO1.1: The specification for the drone delivery system contains a diverse set of requirements, including functional, measurable, and qualitative/measurable requirements. This diversity drives the need for multiple testing strategies, due to the differing natures of the requirements being tested

Functional requirements define the core behaviour of the system, in this case drone movement, precision, strict avoidance of no fly zones and assignment of dispatch records to drones based on capability. In terms of movement, drones must move in a strict step size of 0.00015 degrees, and must only move in one of sixteen compass directions. Precision rules determine whether a target is close enough to be considered reached, with closeness being defined as a circular tolerance of 0.00015 degrees. Medical dispatch records will be assigned to drones based on capabilities necessary for each delivery, such as requiring heating or cooling, and finally drones must avoid restricted areas, never crossing or entering no fly zones under any circumstances.

Measurable quality attributes introduce clear pass/fail criteria. These include correctness of cost calculation (determined as total cost for flight = constants for take-off and return + moves on path * cost per move), efficiency constraints (minimising total cost and moves), and a hard performance constraint that states tasks must be completed within 30 seconds. These requirements are well suited for automated checking and performance measurement due to their clear pass/fail criteria.

Qualitative and dependability requirements are expressed implicitly through safety, robustness, and reliability expectations. Examples include ensuring that drones never violate safety constraints, behave predictably under edge cases like blocked paths, and reliably return to a valid state. These requirements motivate negative testing and fault-focused tests rather than simple output checking.

LO1.2: These requirements apply at multiple levels of the system, enabling different forms of verification and limited validation.

At the unit level, requirements concern deterministic and local behaviour. Examples include validating compass angle constraints, fixed step size movement, geometric calculations for distance and region membership, and precision tolerance checks. These are well suited to unit level verification tests because inputs and expected outputs are precisely defined.

At the integration level, requirements concern interactions between components. Examples include combining validation, geometry and pathfinding logic, and using ILP endpoint data for drone availability, service points, and restricted areas. Interaction tests verify that independently correct components interact correctly when composed.

At the system level, requirements describe end-to-end behaviour, such as complete delivery route calculations, enforcing performance limits, and correctly computing total cost correctly across multiple drones and dispatch records. These tests provide limited validation-like assurance by exercising realistic workflows, even though the specification defines correctness rather than user behaviour.

LO1.3: Different testing approaches were selected based on requirement type and level.

Deterministic unit testing and boundary value testing are used to verify movement rules, precision thresholds, and validation logic. These approaches are appropriate because the specification defines exact constraints and allowable values.

Negative testing is used extensively to enforce safety properties, particularly no fly zone avoidance and rejection of invalid requests. This is appropriate for safety critical requirements where the absence of bad behaviour is more important than producing a specific output.

Integration testing is used to assess robustness when combining controllers, validation, geometry and mocked external services. This helps detect faults that only appear when components interact.

System level and performance testing is used to verify execution time limits and overall route computation behaviour under realistic synthetic workloads. Instrumentation and time bounded assertions are appropriate given the explicit 30 second performance requirement.

LO1.4: The chosen testing approaches are appropriate for the drone delivery system as the specification defines precise, deterministic behaviour with clear pass/fail criteria, making verification focused testing highly effective. Unit and integration testing provide high leverage in detecting faults in movement logic, safety constraints and data interaction, while system level testing increases confidence in performance and end-to-end correctness.

However, this testing approach is heavily verification driven, and offers limited validation due to the way user needs are entirely given in the specification rather than found through real world usage, leaving little room for scope or usability testing. Given the project constraints and safety critical nature of the system, this is still appropriate and supports a cost effective testing process that focuses on correctness, safety and performance.